

School of Science, Computing and
Emerging Technologies



Unit Outline

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COS30045

Data Visualisation

Semester September 2026

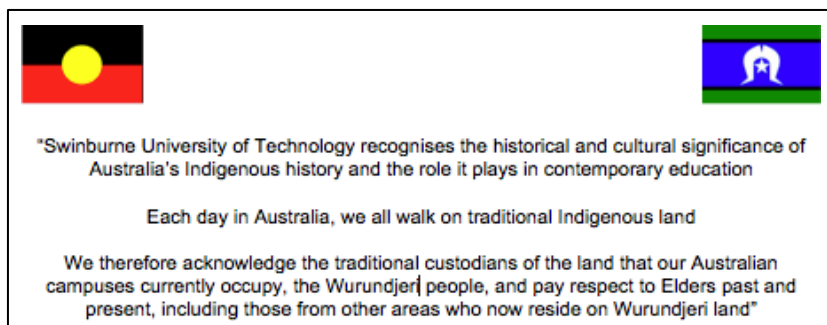
Please read this Unit Outline carefully.

It includes:

PART A Unit summary

PART B Your Unit in more detail

PART C Further information



PART A: Unit Summary

Unit Code(s)	COS30045
Unit Title	Data Visualisation
Duration	12 weeks
Total Contact Hours	36 hours
Requisites:	
Pre-requisites	COS10009 Introduction to Programming
Co-requisites	None
Concurrent prerequisites	None
Anti-requisites	None
Assumed knowledge	None
Credit Points	12.5
Campus/Location	Swinburne Vietnam
Mode of Delivery	Blended
Assessment Summary	Project Assignment Laboratory and Design Exercises

Aims

This unit aims to instruct students on the fundamental principles of information visualisation, and good design practices. It will examine and describe in detail main types of information graphics and visual representations for variety of applications.

Unit Learning Outcomes

Students who successfully complete this unit can:

1. Critically evaluate data visualisations and propose improvements based on an understanding of human perception and cognition and data visualisation design principles.
2. Apply a structured design process to create effective visualisations.
3. Conceptualise and iterate data visualisation designs using sketching and low fidelity prototyping techniques.
4. Create web-based interactive data visualisations using a real-world data set.

Graduate Attributes

The Swinburne Graduate Attributes describe the capability of our graduates to use knowledge, skills and behaviours to contribute to society meaningfully and positively. They include professional, self-directed learning and future-ready skills. This unit contributes to the development of the following Swinburne Graduate Attributes:

GA2 Communication using different media

- GA5 Information literacy

Other graduate attributes may be practised in the unit but are not formally taught as part of the unit content, nor incorporated within formal assessment.

Content

- Introduction to data visualisation
- Brief history of data visualisation
- Data visualisation design guidelines, graphical integrity
- Visual variables: marks and channels
- Visualisation critique
- Analysis of user tasks in visualisation usage
- Data sets and types
- Interaction: Manipulating view, Filtering
- Arranging Tables, Graphs and Maps
- Colour theory
- Human perception and cognition
- User research
- Introduction to Data Driven Documents (D3)

PART B: Your Unit in more detail

Unit Improvements

Feedback provided by previous students through the Student Survey has resulted in improvements that have been made to this unit. Recent improvements include:

Addition of content and exercises about data processing
Lecture content and tutorial content better aligned

Unit Teaching Staff

Name	Role	Location	Email / Teams	Consultation Times
Mr. Vu Ngoc Binh	Lead Coordinator	Ha Noi	bnvu@swin.edu.au	By email appointment
Mr. Quang Chiem	Unit Coordinator	HCM City	qchiem@swin.edu.au	By email appointment
Mr. Quach Huu Nghia	Unit Coordinator	Can Tho	hquach@swin.edu.au	By email appointment

Learning and Teaching Structure

Category	Activity	Total Hours	Hours per Week	Teaching Period Weeks
Live Online	Lecture	12 hours	1 hour	Weeks 1 to 12
On-campus	Class	24 hours	2 hours	Weeks 1 to 12
Online	Directed Online Learning and Independent Learning	18 hours	1.5 hours	Weeks 1 to 12
Unspecified Activities	Independent Learning	96 hours	8 hours	Weeks 1 to 12

Week by Week Schedule

Wk	Date*	Lecture Content	Class Task	Assessment
0	No classes	Self guided	L00: Set up environment	
1	14 Sep	Introduction to Data Visualisation Data processing	L01: Data processing	
2	21 Sep	Design guidelines, visual variables, , chart types	L02: Data exploration	
3	28 Sep	Story Telling Data sources and privacy	L03: Charts for communication	
4	5 Oct	Introduction to D3 D3 Basics	L04: Introduction to D3	Demonstration 1
5	12 Oct	Static visualisations	L05: Common Chart types	Assignment
6	26 Oct	Interactive visualisations	L06: Interactive web visualisations	
7	2 Nov	Visual perception	L07: Visualisation design	Demonstration 2
8	9 Nov	Accessibility	L08: Project Kick-off	Team formation
9	16 Nov	Usability evaluation methods	L09: Usability evaluation	
10	23 Nov	Interaction design	L10: Project Consultation	Project Stand-up 1
11	30 Nov	Data visualisation design	L11: Project Consultation	Project Stand-up 2
12	14 Sep	Review	L12: Project Consultation	Project Stand-up 3

*Week beginning

Assessment

a) Assessment overview

Tasks and Details	Individual or Group	Weighting	Mapped Unit Learning Outcomes	Mapped Graduate Attributes	Assessment Due Date
Demonstrations	Individual	15%	1, 2, 4	GA2, GA5	Weeks 4,7
Assignment	Individual	10%	1		Week 6
Project					
Stand-ups	Individual	15%	2, 3, 4	GA2, GA5	Week 10, 11 and 12
Process Book and website	Team	40%	1, 2, 3, 4	GA2, GA5	Week 12
Reflection	Individual	20%	1, 2, 3, 4		Week 12

Assessment Requirements	Details
b) Use of generative AI (genAI) in this unit	Generative AI must not be used for any assessment in this unit. All assessments must be completed entirely without genAI assistance. This includes drafting, paraphrasing, editing, or generating images/code. Misuse will be addressed under Swinburne's academic integrity processes.
c) Hurdle requirements	To pass this unit, you must: achieve an overall mark for the unit of 50% or more
d) Final assessment period	If the unit you are enrolled in has a final assessment (including invigilated exams), you will be expected to be available for the entire final assessment period including any Special Exam period.
e) Submission requirements	Assignments and other assessments are generally submitted online through the Canvas assessment submission system which integrates with the Turnitin. Please ensure you keep a copy of all assessments that are submitted. In cases where a hard copy submission is required an Assessment Cover Sheet must be submitted with your assignment. The standard Assessment Cover Sheet is available from the Submitting work webpage or www.swinburne.edu.au/studentforms/

f) Extensions, late penalties and special consideration	<u>Extensions</u> For extensions (other than those granted through Special Consideration), please contact your Unit Coordinators who will assess on a case-by-case basis.
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	<u>Late Penalties</u> Unless an extension has been approved, late submissions will result in a mark penalty. You will be penalised 5% of the available mark for each calendar day the task is late, up to a maximum of seven calendar days. After seven calendar days, a zero mark will be recorded. For example: • if you achieve 90/100 on an assessment task but the task was submitted two days late, a late penalty of 10 is applied, and your mark is recorded as 80/100. • if you achieve 29/35 on an assessment but the task was submitted five days late, a late penalty of 8.75 (25% of the 35 available marks) is applied, and your mark is recorded as 20.25/35. <u>Special Consideration</u> If your studies have been adversely affected by serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you can apply for special consideration. Applications for Special Consideration are made via an online form <u>no later than 5.00pm</u> of the third working day after the assessment due date. For further information about eligibility and how to apply, see https://www.swinburne.edu.au/life-at-swinburne/student-support-services/special-consideration-assistance/
g) Referencing	To avoid breaching academic integrity, you are required to provide references whenever you include information from other sources in your work and acknowledge when you have used Artificial Intelligence (AI) tools (such as Microsoft Copilot). Further details regarding academic integrity are available in Section C of this document. Referencing conventions required for this unit are: APA7 Helpful information on referencing can be found at http://www.swinburne.edu.au/library/referencing/

h) Groupwork guidelines	A group assignment is the collective responsibility of the entire group, and if one member is temporarily unable to contribute, the group should be able to reallocate responsibilities to keep to schedule. In the event of longer-term illness or other serious problems involving a member of group, it is the responsibility of the other members to notify immediately the Unit Convenor or relevant tutor. All group members must be satisfied that the work has been correctly submitted. Any penalties for late
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	submission will generally apply to all group members, not just the person who submitted.
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b) Final unit results

a) Supplementary assessment	Students will be notified by the University if they are eligible for supplementary assessment. Students are offered supplementary assessment in this unit where they have: - failed the unit and the final mark for the unit is between 45 and 49, and - attempted all assessment tasks during the teaching and final assessment periods. Students are not eligible for supplementary assessment if they have received a mark reduction because of an academic misconduct finding under the <i>Student Academic Misconduct</i> regulations. Students may only undertake supplementary assessment in one unit per teaching period. Where students are eligible for supplementary assessment in multiple units in a teaching period, they may select which unit they will undertake supplementary assessment in. If students pass the supplementary assessment task, their final result for the unit will be changed to a Passed Supplementary (SP) with a mark of 50%. Information on supplementary assessment is available on the Swinburne website.
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Required Textbook(s)

The required textbook(s) can be purchased from bookshops and may be available through the Swinburne Library.

Dufour, A-M., & Meeks, E. (2024). D3. js in Action. Manning Publications.

Knaflic, C. N. (2015). Storytelling with data: A data visualization guide for business professionals. John Wiley & Sons.

Recommended Reading Materials

Swinburne Library has a large collection of resources. Listed below are some references that will provide valuable supplementary information to this unit. It is also recommended that you explore other sources to broaden your understanding.

See Canvas for list of readings

PART C: FURTHER INFORMATION



For further information on any of these topics, refer to Swinburne's Student webpage <http://www.swinburne.edu.au/student/>

Student behaviour and wellbeing

All students are expected to: act with integrity, honesty and fairness; be inclusive, ethical and respectful of others; and appropriately use University resources, information, equipment and facilities. All students are expected to contribute to creating a work and study environment that is safe and free from bullying, violence, discrimination, sexual harassment, vilification and other forms of unacceptable behaviour.

The [Student Charter](#) describes what students can reasonably expect from Swinburne in order to enjoy a quality learning experience. The Charter also sets out what is expected of students with regards to your studies and the way you conduct yourself towards other people and property.

You are expected to familiarise yourself with University regulations and policies and are obliged to abide by these, including the [Student Academic Misconduct Regulations](#), [Student General Misconduct Regulations](#) and the [People, Culture and Integrity Policy](#). Any student found to be in breach of these may be subject to disciplinary processes.

Examples of expected behaviours are:

- conducting yourself in teaching areas in a manner that is professional and not disruptive to others
- following specific safety procedures in Swinburne laboratories, such as wearing appropriate footwear and safety equipment, not acting in a manner which is dangerous or disruptive (e.g. playing computer games), and not bringing in food or drink
- following emergency and evacuation procedures and following instructions given by staff/wardens in an emergency response.

Canvas

You should regularly log on to the Swinburne learning management system, Canvas. You can access Canvas via the [Student login](#) webpage or <https://swinburne.instructure.com/>. Canvas is updated regularly with important unit information and communications.

Communication

All communication will be via your Swinburne email address. If you access your email through a provider other than Swinburne, then it is your responsibility to ensure that your Swinburne email is redirected to your private email address.

Academic Integrity

Academic integrity is about taking responsibility for your learning and submitting work that is honestly your own. It means acknowledging the ideas, contributions and work of others; referencing your sources and acknowledging the use of generative artificial intelligence;

contributing fairly to group work; and completing tasks, tests and exams without cheating. Artificial intelligence tools should only be used where approved by the Unit Coordinators.

Swinburne University uses the Turnitin system, which helps to identify inadequate citations, poor paraphrasing and unoriginal work in assignments that are submitted via Canvas. Your Unit Coordinators will provide further details.

Plagiarism, collusion, contract cheating, unauthorised file sharing, falsification, fabrication, manipulation or misrepresentation of information, reuse of previous work and noncompliance with instructions in an invigilated or non-invigilated assessment item are all breaches of academic integrity and treated as academic misconduct. Examples of breaches of academic integrity include, but are not limited to:

- submitting work as your own for assessment that has been fully or partially completed by a third party, either paid or unpaid
- using output from artificial intelligence tools (e.g. ChatGPT) in whole or part without acknowledgement and/or without the approval of the Unit Coordinators
- using another person's work or ideas as though it is your own work, without appropriate attribution
- working closely with another student or group of students (either past or current), to submit for assessment, some or all of the other student or students' work as your own work
- sharing without permission of the Unit Coordinators, Swinburne resources or other material related to assessment to an entity or document repository site
- creating, intentionally modifying or inventing information that is intended to be submitted as part of an assessment item
- using the whole or part of a computer program written by another person as your own without appropriate acknowledgement
- poorly paraphrasing somebody else's work
- using a musical composition or audio, visual, graphic and photographic work created by another person without acknowledgement
- enabling others to cheat, including letting another student copy your work or by giving access to a draft or completed assignment
- letting someone or something else impersonate you, or you impersonate someone else in an invigilated or non-invigilated assessment item
- accessing, obtaining and/or providing to others unauthorised materials relating to an invigilated or non-invigilated assessment item.

The penalties for academic misconduct can be severe, ranging from a zero grade for an assessment task through to exclusion from Swinburne. For further details, see <https://www.swinburne.edu.au/student-login/academic-integrity/>

Student support

Swinburne offers a range of services and resources to help you complete your studies successfully, such as study and learning support and health and wellbeing services. Your Unit Coordinators or studentHQ can provide information about the study support and other services available for Swinburne students. For further information, see the [Student support services](#) web page.

Special consideration

If your studies have been adversely affected due to serious and unavoidable circumstances outside of your control (e.g. severe illness or unavoidable obligation), you may be able to apply for special consideration (SPC).

Applications for Special Consideration will be submitted no later than 5.00pm on the third working day after the submission/sitting date for the relevant assessment component. Please contact your Unit Coordinator, Lecturer or Tutor for more information on how to apply.

If you are studying at Hawthorn, please refer to <https://www.swinburne.edu.au/life-at-swinburne/student-support-services/special-consideration-assistance/>

Otherwise, please contact your local StudentHQ/student support services team for more information.

Note: Submitting fraudulent (fake or altered) medical certificates is considered misconduct and can lead to serious penalties from Swinburne. In addition, your doctor may report fraudulent medical certificates as a prosecutable offence under the Victorian Crimes Act.

AccessAbility Services

If you are a student with a disability, medical or mental health condition or you have significant carer responsibilities, you may require reasonable adjustments to fully access and participate in education. Swinburne's AccessAbility Services can develop an Education Access Plan (EAP) that includes the services and reasonable adjustments that you need.

It is recommended that you register with AccessAbility Services when you first commence your course, but you can contact the service at any time during your studies to find out about reasonable adjustments. Contact [Accessibility Services](#) to discuss further.

Once you have an approved EAP, you should contact the Unit Coordinator as early as possible (e.g. beginning of semester) to discuss your adjustments so that we can better support your learning. Otherwise, please contact your local StudentHQ/student support services team to find out more.

Review of marks

An independent marker reviews all fail grades for major assessment tasks. In addition, a review of assessment is undertaken if your final result is between 45 and 49 or within 2 marks of any grade threshold.

You can ask the Unit Coordinators to check the result for an assessment item or your final result. Your request must be made in writing within 10 working days of receiving the result. The Unit Coordinators can discuss the marking criteria with you and check the aggregate marks of assessment components to identify if an error has been made. This is known as local resolution. If you are dissatisfied with the outcome of the local resolution, you can lodge a formal complaint.

Feedback, complaints and suggestions

In the first instance, discuss any issues with your Unit Coordinators. If your concerns are not resolved or you would prefer not to deal with your Unit Coordinators, then you can complete a feedback form. See <https://www.swinburne.edu.au/corporate/feedback/>

Advocacy

Should you require assistance with any academic issues, University statutes, regulations, policies and procedures, you are advised to seek advice from an Academic Student Support Officer at Swinburne Student Life.

For an appointment, please call 19006412 or lodge a query via Student Portal at <https://portal.swin.edu.vn>